

XBPF DESIGN AND PROTOTYPING

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Abstract

The CERN Beam Instrumentation Group has developed a new scintillating fibre beam profile monitor for the secondary beam lines of the CERN North Experimental Area. This innovative monitor employs plastic scintillating fibres, read out with silicon photomultipliers, to provide a cost-effective and efficient solution for beam profile measurement. The design goals for the new monitor included ease and low cost of production, achieving a particle detection efficiency above 95 %, compatibility with beam intensities ranging from 1 to 10^8 particles per second, a spatial resolution of 1 mm, a low material budget, coverage of an active area of $10\text{ cm} \times 10\text{ cm}$ or $20\text{ cm} \times 20\text{ cm}$, operability in a vacuum environment, and equipped with in/out motorisation for retracting the equipment from the beamline. A prototype was tested at the CERN East and North Area facilities, demonstrating excellent performance and validating the design for mass production.

INTRODUCTION

CERN is undertaking a consolidation project of the North Area complex, the largest experimental area at CERN and home to a rich program of fixed-target physics. As part of this effort, the beam instrumentation is also being upgraded. The secondary beamlines are still operating with the original instrumentation, which was designed and installed in the 1970's. Despite their longevity and maintenance, these instruments are now showing signs of degradation due to ageing and radiation exposure, and spare parts are no longer available. At present profile monitoring is done using Multi-Wire Proportional Chambers [1], Delay Wire Chambers [2] and the Filament Scintillator Scanner (FISC) [3]. The detector of choice to replace the existing monitors is the scintillating fibre monitor (XBPF) developed by the Beam Instrumentation (BI) Group [4]. The new detectors are designed for operation in vacuum environments, as the beamlines will be in vacuum wherever possible to improve beam quality, and feature retractable mechanisms to minimize material budget on the line when not in use and extend fibre lifespan. Two types are needed one covering an area of $10\text{ cm} \times 10\text{ cm}$ and another one covering an area of $20\text{ cm} \times 20\text{ cm}$. This paper presents the design and validation of the $10\text{ cm} \times 10\text{ cm}$ prototype, along with ongoing development efforts.

XBPF DETECTOR DESIGN

The overall detector head design has already been implemented in secondary beamlines at CERN [4]. Earlier versions operated in vacuum but were fixed in place, requiring manual intervention for repositioning. The new design

introduces an in/out motorisation system that allows the detector to be retracted from the beam, improving operational flexibility. The fibres selected have a thickness of 1 mm and a square cross-section, which enhances detector homogeneity. In practice, 96 fibres are used to match the readout electronics, which have 96 channels. As a result, the actual active area is $9.6\text{ cm} \times 10\text{ cm}$. To streamline development, existing vacuum chambers from the FISCs were repurposed, which required adherence to predefined flange and geometry constraints.

In-Out Mechanism

The detector is to be placed in a primary vacuum 10^{-3} mbar environment. A key challenge was minimizing vacuum forces acting on the actuator mechanism. Fibre routing to the readout board required careful consideration of bending radii, requiring a compact actuator footprint. Early concepts involving internal readout boards were discarded due to the need for active cooling in vacuum. The most practical solution for in-out movement was the use of edge-welded bellows. Table 1 summarizes the vacuum forces for available bellow configurations..

Table 1: Vacuum Forces for Different Bellow Sizes

Bellow Size	Vacuum Force
DN100	1121 N
DN150	2287 N
Racetrack 40/130	823 N

The racetrack-shaped bellow was selected for its compact footprint and reduced vacuum force, which eased integration and minimized mechanical stress. A pneumatic actuator was chosen for its simplicity—requiring only ON/OFF control signals—and adjustable force via actuator size or supplied pressure. A 150 mm stroke ensures full beam clearance when retracted. The mechanism is illustrated in Fig. 1.

Detector Head

The detector head comprises three main components:

- Interface flange: Seals the vacuum and connects to the readout board.
- Active area: Interacts with the beam.
- Support structure: Ensures mechanical stability and alignment.

The overall dimension of the support structure inside the vacuum was restricted by the choice of the vacuum bellow. To avoid modifications to the read out board the same fibre pattern was used on the interface flange as on previous XBPFs produced. This meant that we had to start the design

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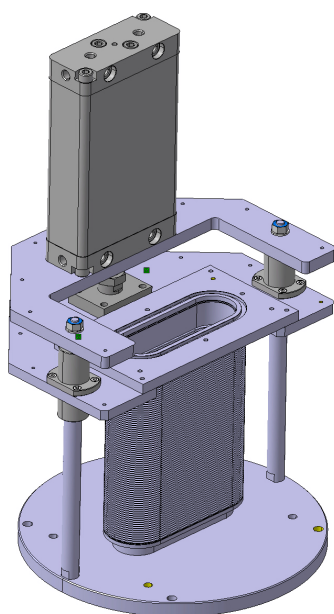


Figure 1: The designed mechanism, showing the flange interfaces, the racetrack edge welded bellow and the pneumatic actuator.

with several limitations. Figure 2 shows the main components of the detector assembly. The parts shown in dark grey are made out of black Polyoxymethylene (POM). Dark materials have to be used to avoid parasitic light reflections. Additionally, to avoid static charges on the fibre which makes it difficult to put them next to each other, all the fibres are coated with an aluminium coating which also has the benefit to reduce optical crosstalk between fibres. The structure in light grey is made out of aluminium which is used to ensure stiffness for the detector. The active area consist of two POM halves that clamp the fibres in position. The POM support themselves are fixed to the aluminium support. Some pins allow the detector to be moved along the length of the support which allows us to align the centre fibre to the nominal position.

Read Out Box

The read out board also had to be modified slightly due to the constraints given by the mechanism Fig. 1. Core components remain consistent with fixed XBPFs. Light from the fibres is detected by silicon photomultipliers (SiPMs), with optical grease ensuring optimal contact. Additional fixing points were added to the interface flange and readout board to maintain alignment. A new interface board and light-tight enclosure were designed to shield the SiPMs from external light, ensuring accurate signal acquisition.

PROTOTYPE PRODUCTION

As shown in Fig. 2 a small groove was machined into the aluminium plate to ensure that the fibres remain securely positioned after being fed through the slot and inserted into the interface flange. This design feature significantly reduces

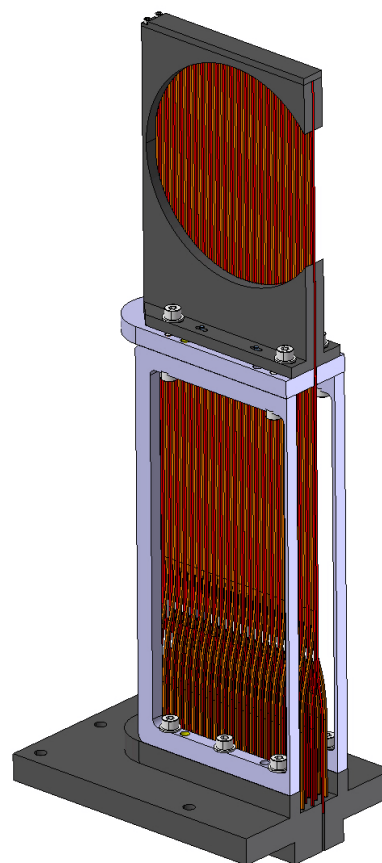


Figure 2: The design of the detector head with a cut view.

the risk of fibre misalignment or mixing. The fibres were then inserted into the hole pattern of the intermediate flange. These holes were intentionally made oversized to facilitate insertion. Once all fibres were correctly positioned, optical glue was applied to the bottom of the flange to seal the holes and ensure vacuum tightness. Figure 3 shows the first two prototypes following the initial fibre gluing process. The next step involved trimming the fibres to their final length. To achieve optimal optical performance, both ends of the fibres were polished, the first iteration was polished by hand on the active area side where as the interface flange was done using a diamond cutter mounted on a milling machine. For the latest iteration the polishing was done on both sides by with the milling machine. This polishing process also removes a small amount of support material, which must be accounted for in the design phase.

Validating the Prototype in the Lab

Once assembled, the prototype underwent a series of validation tests:

- **Vacuum Integrity:** Leak testing confirmed the assembly was vacuum-tight.
- **Mechanical Movement:** The in-out mechanism was tested for smooth operation.

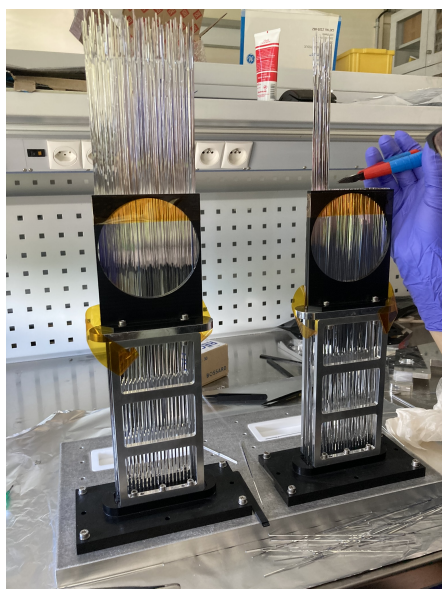


Figure 3: Prototype production of detector head, trimming the length of the fibres once glued into the interface flange.

- **Particle Detection:** Using a Strontium-90 radioactive source, the fibre arrangement and the optical coupling between the fibres and the SiPMs were verified. The detection efficiency, a key performance parameter, was also measured.
- **Durability Testing:** The mechanism was cycled 1000 times with the readout board attached to ensure mechanical reliability and resistance to vibration-induced wear.

BEAMLINE TESTING

Following successful lab validation, the prototype was installed in the H8 beamline of the CERN North Experimental Area. It was positioned downstream to existing profile monitors to enable direct performance comparison. Figure 4 shows a direct comparison of the XPBF prototype with a Delay Wire Chamber. The XPBF provides more accurate profiles than the Delay Wire Chamber. During these tests the XPBF showed a detection efficiency of 96 %, exceeding the required performance specifications.

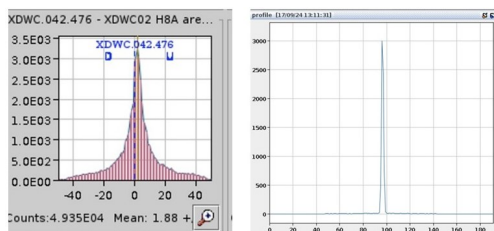


Figure 4: Profile of a 10^6 particles per beam extraction of a hadron beam in the North Area measured by a Delay Wire Chamber (left) and the XPBF (right) – the XPBF produces more accurate profiles.

OUTLOOK

Following the successful prototype validation, preparations are underway for series production of the first batch of 20 XPBF instruments. Development is also progressing on a larger $20\text{ cm} \times 20\text{ cm}$ detector variant. This design introduces new challenges due to increased vacuum forces—estimated at 2075 N—even with a suitable racetrack-shaped edge-welded bellow. The larger bellow also requires additional structural support. Unlike the $10 \times 10\text{ cm}$ version, this design will include a custom vacuum chamber, offering greater flexibility to address these challenges. Design work is ongoing to resolve the new technical constraints.

CONCLUSION

A robust and vacuum-compatible design has been successfully developed for the XPBF detector, enabling reliable in-out movement and high-resolution beam profiling. Extensive laboratory and beam tests in the CERN North Experimental Area have validated its performance and confirmed its suitability as a replacement for ageing instrumentation. Series production of the first batch is underway, with installation planned during CERN's Long Shutdown 3 (2026-2029). Further development is focused on scaling the design to accommodate larger active areas while preserving the same operational principles and performance standards.

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